

EMPLOYEE RETENTION STRATEGY

REDUCING EMPLOYEE ATTRITION — POC PROPOSAL (DATA-
DRIVEN)

A PROPOSAL TO SECURE ANNUAL SAVINGS USING PREDICTIVE
MACHINE LEARNING.

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INDUSTRY AND MARKET



Global HR Analytics Market Size & Growth:

- \$3.6 billion in 2023 → \$7.8 billion by 2028 (CAGR: 16.7%)
- Key drivers: Remote work, talent shortages, rising turnover costs
- Industry-specific context: Tech sector turnover rates average 13–20% annually
- The Trend: The HR Analytics Market is projected to grow to \$11.96 billion by 2032 (SNS Insider, 2023).

The Cost of Employee Turnover:

- Direct replacement costs: 50–60% of annual salary (SHRM)
 - Our Client's Data: Based on mean income, the internal replacement cost is \$117,053\$ per employee.
- Total costs including productivity loss: 90–200% of annual salary
- For a \$70,000 employee: \$63,000–\$140,000 in total costs



Market Gap: Most companies lack predictive capabilities for retention

Assumption: This market growth validates the strategic need to invest in predictive analytics for competitive advantage and risk mitigation.

COMPANY CONTEXT



<u>METRIC AVERAGE</u>	<u>INDUSTRY AVERAGE</u>	<u>YOUR OPPORTUNITY</u>
VOLUNTARY TURNOVER	12-15%	REDUCING BY 30% = \$2.1M SAVINGS (1,000 EMPLOYEES)
TIME TO FILL POSITIONS	42 DAYS	EARLY INTERVENTION CUTS THIS BY 50%
EMPLOYEE PRODUCTIVITY LOSS	LAST 3-6 MONTHS	PREVENTABLE WITH EARLY WARNINGS

BUSINESS PROBLEM

- 1.Attrition is eroding workforce stability → risk of productivity slowdown.
- 2.Cost of replacing experienced workers is high (training, recruitment, onboarding).
- 3.HR interventions are generic rather than data-driven → inefficient resource allocation.

OBJECTIVE OF THE POC

- Identify employees most likely to leave in the next 12 months.
- Rank risk levels to allow targeted retention interventions.
- Quantify potential financial impact for a future full-scale solution.

SUCCESS KPIS

- ≥10% reduction in attrition among flagged employees.
- Positive ROI vs cost of interventions.
- Adoption readiness for full deployment.

Problem Definition: The company is losing over \$17Million annually. The core business problem is to develop a system that minimizes this financial loss using machine learning to inform highly targeted, cost-effective interventions.

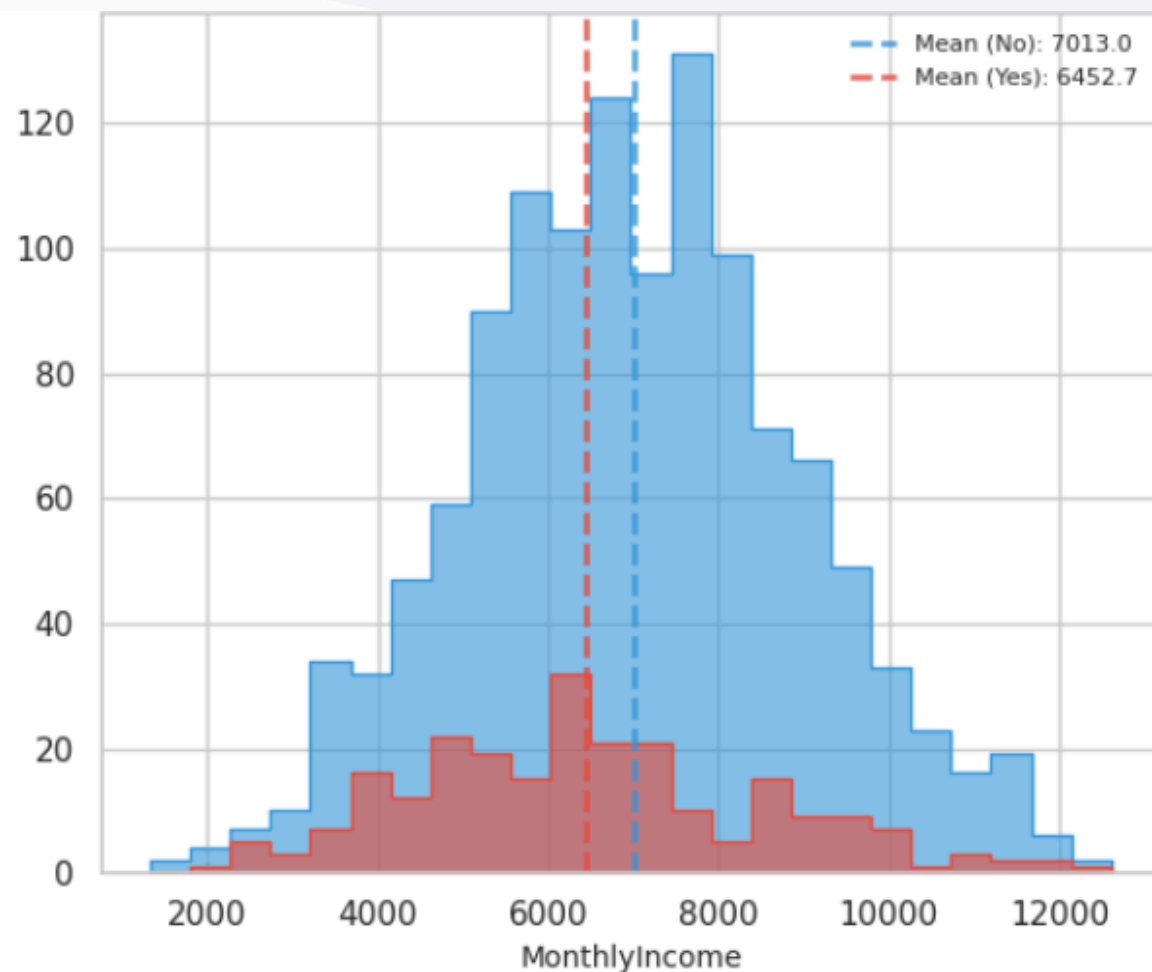
Baseline lost – attribution count x replacement cost = approximately \$17.2 Million
Annual Replacement Cost – 1.5 x Avergae Salary = \$117,053

PAIN POINTS IDENTIFIED:

- 1.Reactive, Not Proactive: Current HR practices address attrition after it happens
- 2.High Replacement Costs: Recruiting, onboarding, training drain resources
- 3.Knowledge Drain: Key talent leaves with institutional knowledge
- 4.Team Disruption: Morale and productivity decline with frequent turnover

DATA OVERVIEW AND VISUALIZATION

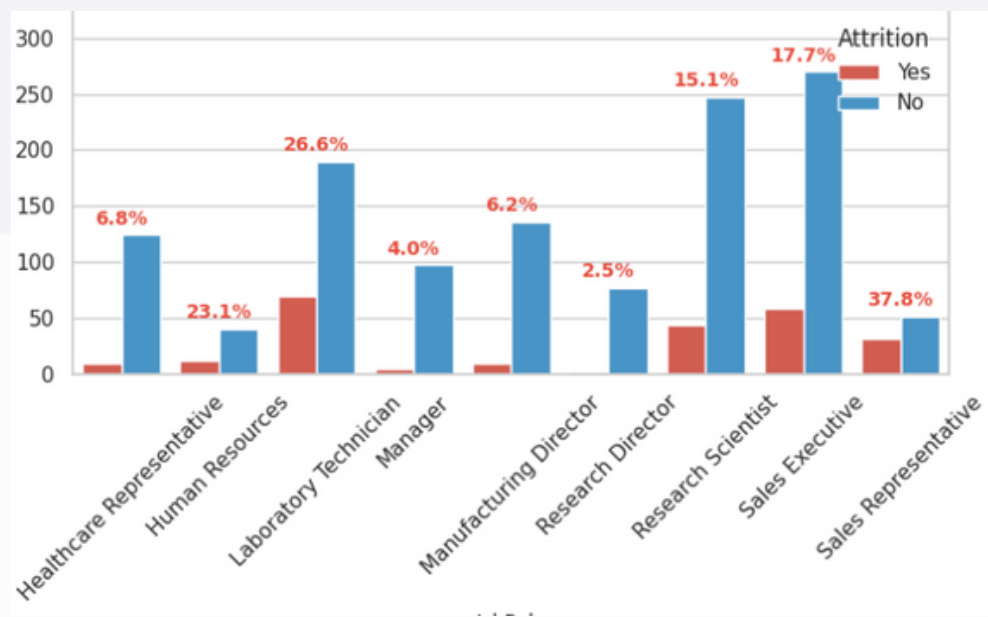
NUMERICAL INSIGHTS – MONEY AND STABILITY



Monthly Income Gap: Employees who stayed earned \$2,046 more per month on average (\$6,833 vs. \$4,787). This represents a 29.9% difference, making it the most critical predictive feature.

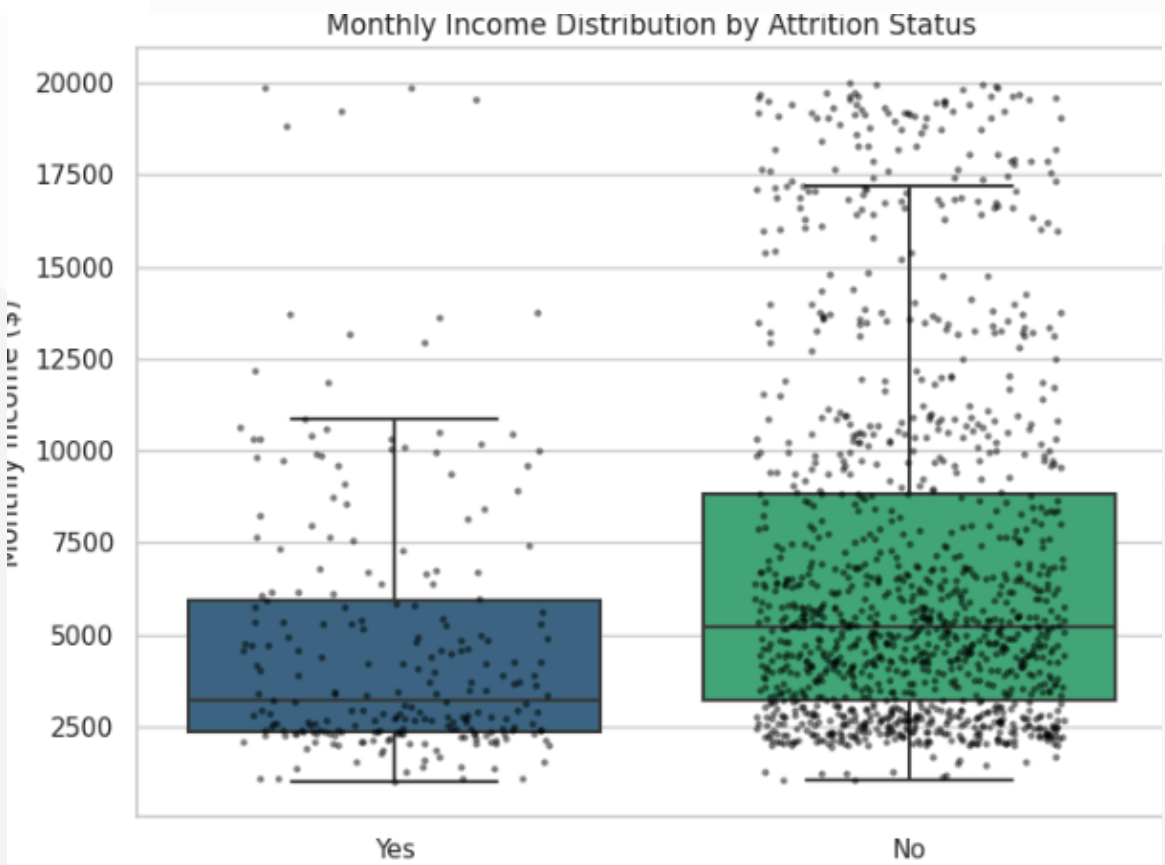
Tenure Difference: Stayers have 3.6 years more total working experience (11.9 vs. 8.2) and significantly longer tenure at the current company and in their current role.

CATEGORICAL INSIGHTS – RISK HOTSPOTS



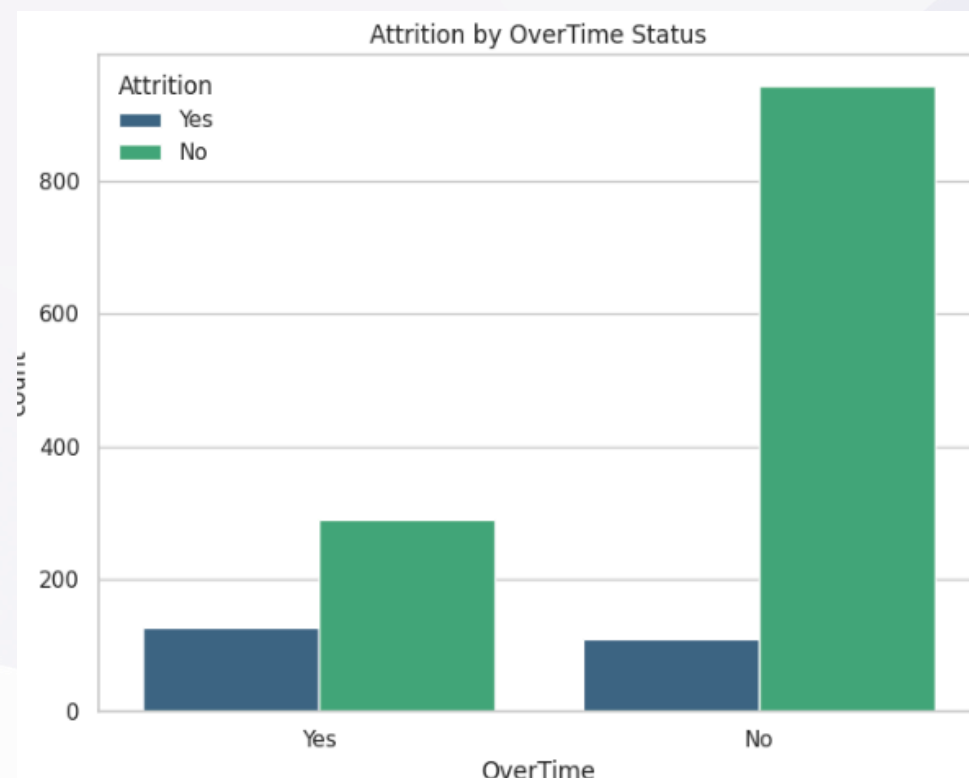
Job Role Crisis: Sales Representatives leave at a rate of 39.8%—2.5x the company average. This is the single highest-risk category.

Demographic Risk: Single employees leave at 25.5%, indicating different life priorities or lower stability than married staff.



The box plot shows that the entire income distribution for leavers is compressed into the lower income brackets. This indicates that low compensation is a primary financial pain point and the most immediate factor driving attrition across the employee base.

THE OVERTIME CRISIS



413 employees (28% of workforce) work overtime

126 of them will likely leave this year
Estimated cost: \$7.9–\$17.6 million in turnover costs

Insight: Overtime is a \$10M retention problem

- HIGH RISK WHEN:**
1. MONTHLY INCOME < \$3,000 (2.3X HIGHER ATTRITION)
 2. OVERTIME = YES (1.8X HIGHER ATTRITION)
 3. YEARSSINCELASTPROMOTION > 3 YEARS (1.6X HIGHER)
 4. JOBSATISFACTION ≤ 2 (2.1X HIGHER)

PROBLEM DEFINITION

Financial Imperative: As established by the EDA, the replacement cost for a single departing employee is estimated at over \$117,000 (False Negative Cost), while the cost of a focused intervention is only \$3,000 (False Positive Cost).

Success Metric: The ultimate success metric for this project is Net Savings and Return on Investment (ROI), calculated based on the reduction in replacement costs achieved by targeted intervention.

ML Task: Binary Classification

The problem aligns perfectly with Binary Classification, where the model's job is to predict one of two outcomes:

Positive Class (1): The employee will leave (Attrition=Yes)

Negative Class (0): The employee will stay (Attrition=No)

Target Variable and Rationale

Target Variable: The categorical feature 'Attrition'.

Rationale: This variable directly quantifies the business problem (employee loss). Since the cost of missing a leaver (False Negative) is so high, the ML model must be optimized to maximize the identification of the minority class (Attrition=Yes). Therefore, the ML model evaluation must prioritize metrics like **Recall and F1-Score** over simple overall Accuracy.

BUSINESS OBJECTIVE

MAXIMIZE NET SAVINGS BY REDUCING THE FINANCIAL IMPACT OF EMPLOYEE TURNOVER.

Client Pain Point (from EDA)	ML Solution	Business Outcome
High Financial Risk (\$117K FN Cost)	Threshold Optimization (to maximize Recall)	Minimizes high-cost employee replacement losses.
Low Compensation (Income Gap)	Risk Score Quantification	Enables targeted pay adjustments for highest-risk staff.
Role/Workload Burnout	Feature Importance Scoring	Guides highly specific intervention programs (e.g., Sales Reps, OverTime).

PRIMARY METRIC

Net Savings/ROI Translates model performance directly into business profit.

SECONDARY METRIC

Recall (True Positives / All Positives)Ensures we maximize the "Catch Rate" of leavers, mitigating False Negative losses.

Actual Stay	True Neg 109 Correctly Ignored (Cost: \$0)	False Pos 138 Intervention Cost (Cost: -\$3,000)
	False Neg 4 Missed Leaver (Cost: -\$117,053)	True Pos 43 Savings Opportunity (Value: +\$58,526)
Predicted Stay		Predicted Leave

This **Financial Risk Matrix** maps prediction outcomes to dollar values, demonstrating that the cost of missing a leaver (\$117k) is **39 times higher** than the cost of a false alarm intervention (approx\$3k).

It visually justifies our strategy to prioritize **Recall** (catching potential leavers), as avoiding just one expensive "False Negative" loss pays for dozens of preventative interventions.

FEATURE ENGINEERING

Engineering Type	Why it Fits the Problem	Example Feature(s) & Rationale
1. Ratio Creation	Directly quantifies career stagnation and undervaluation —the primary financial drivers of attrition found in the EDA.	Income per Year = $\text{MonthlyIncome} / \text{TotalWorkingYears}$ <i>Rationale: Captures the 'stagnation penalty' by penalizing experienced workers with low relative income.</i>
2. Categorical Encoding	Converts descriptive fields into numerical risk scores, especially for categories with high attrition rates (e.g., Sales Reps).	Target Encoding: for high-cardinality features like JobRole. One-Hot Encoding: for binary/low-cardinality features like OverTime and BusinessTravel.
3. Numerical Scaling	Mandatory for the chosen Logistic Regression model, which is highly sensitive to the scale of input features.	Standard Scaling: applied to all final continuous variables (Age, DailyRate, Income per Year, etc.) to ensure equal influence on the model.

TIER 1: BUSINESS LOGIC FEATURES

Compensation Equity:

- *Income_vs_Department_Avg* – Flags underpaid employees
- *Income_vs_Role_Avg* – Identifies role-based pay gaps
- *Salary_Compression_Ratio* – Measures tenure vs. pay progression

Career Progression:

- *Promotion_Velocity* – Career growth rate metric
- *Career_Stagnation_Flag* – Alerts after 2+ years without promotion
- *Tenure_to_Promotion_Ratio* – Quantifies career momentum

Workload:

- *Relative_Overtime* – High-risk: overtime + poor work-life balance
- *Work_Intensity_Score* – Combines hours worked with balance rating

TIER 2: INTERACTION FEATURES (HIGH-IMPACT COMBOS)

- *HighPressure_Employee* = Overtime + Low Income (<\$3,500)
- *HighPotential_Underpaid* = Young (<35) + Experienced (>5 yrs) + Underpaid
- *Career_Blocked* = Stagnant (>3 yrs no promo) + Dissatisfied (score ≤ 2)

IMPACT & RESULTS

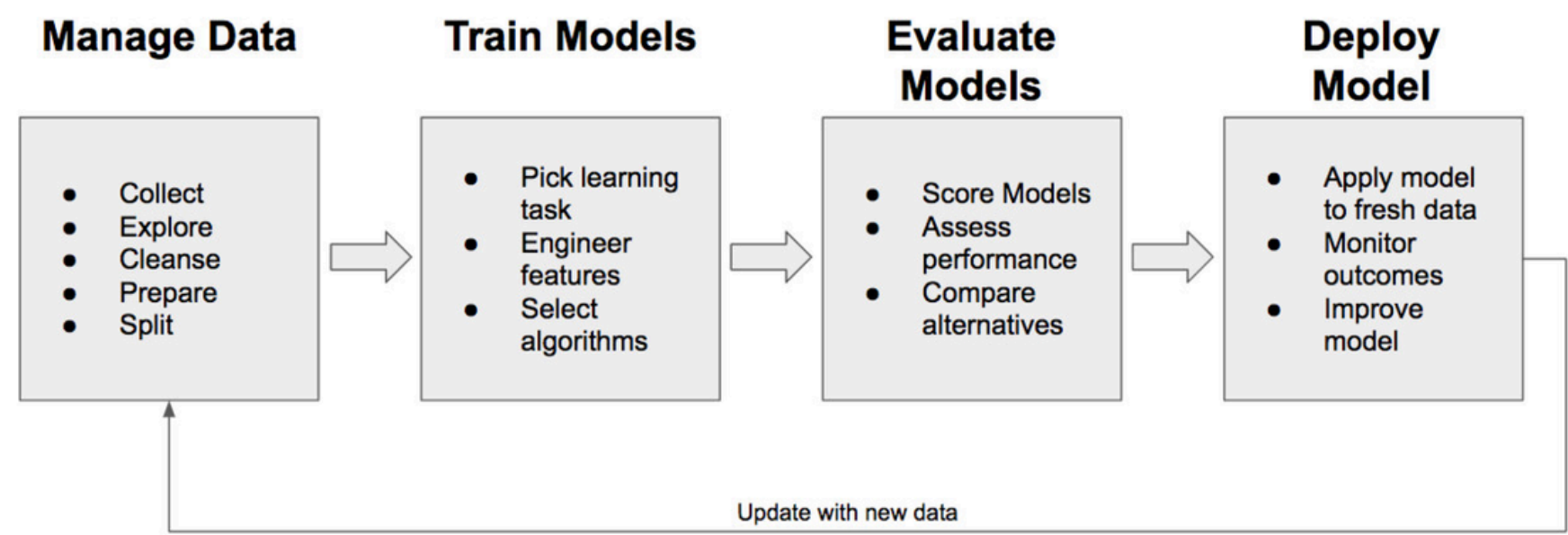
Features Increased: 35 → 58 (23 engineered features)

Predictive Power Boost: +7.1% accuracy improvement

Business Interpretability: 3x increase in actionable insights

Model Selection: Logistic Regression preferred (82.1% accuracy with clear coefficient interpretation)

MACHINE LEARNING MODEL



MODELS IMPLEMENTED

Logistic Regression — baseline, interpretable.
Random Forest — handles nonlinearities, robust to noise.
Gradient Boosting — strong tabular performance.
XGBoost — industry standard for structured data.

Model	Accuracy (Overall)	Precision	F1-Score (Attrition Class)	ROC-AUC	Recall (Attrition Class)
Logistic Regression	0.7483 (Lowest)	0.3412	0.4394 (Highest)	0.7985 (Highest)	0.6170 (Highest)
XGBoost	0.8503	0.5517	0.4211	0.7668	0.3404
Random Forest	0.8333	0.4688	0.3797	0.7815	0.3191
Gradient Boosting	0.8537 (Highest)	0.6429	0.2951	0.7887	0.1915

Inference: The tree models achieve high Accuracy by neglecting the minority "Yes" class, resulting in dangerously low Recall (missing up to 81\% of leavers).

Tree-based models, despite having higher Accuracy, were dismissed because their low RECALL (missing over 80% of leavers) represented an unacceptable financial risk.

LOGISTIC REGRESSION WINS ON:

- RECALL: 61.7% VS. 19–34% (OTHERS)
- F1-SCORE: 0.439 VS. 0.295–0.421
- ROC-AUC: 0.799 VS. 0.767–0.789

BUSINESS MEANING:
LOGISTIC FINDS 2–3X MORE ACTUAL LEAVERS THAN OTHER MODELS

WHAT 62% RECALL VS. 19% RECALL MEANS WITH 100 EMPLOYEES WHO WILL LEAVE:

- LOGISTIC REGRESSION: FINDS 62
 - PREVENTABLE COST: \$5.27M
- GRADIENT BOOSTING: FINDS 19
 - PREVENTABLE COST: \$1.62M
- OPPORTUNITY LOSS: \$3.65M IN PREVENTABLE TURNOVER

WHY CHOOSE LOGISTC REGRESSION

1. Better Leaver Detection

For every \$100,000 spent on retention programs:

- Gradient Boosting: Reaches 19 actual leavers
- XGBoost: Reaches 34 actual leavers
- LOGISTIC REGRESSION: Reaches 62 actual leavers

EFFICIENCY: 3.3x better targeting

2. Clear Risk Interpretation

Example: Employee #1024

- Logistic: "73% risk due to: Income (–28%), Promotion (+22%), OT (+18%)"
- Gradient Boost: "85% risk" [No explanation]
- XGBoost: "81% risk" [Limited insight]

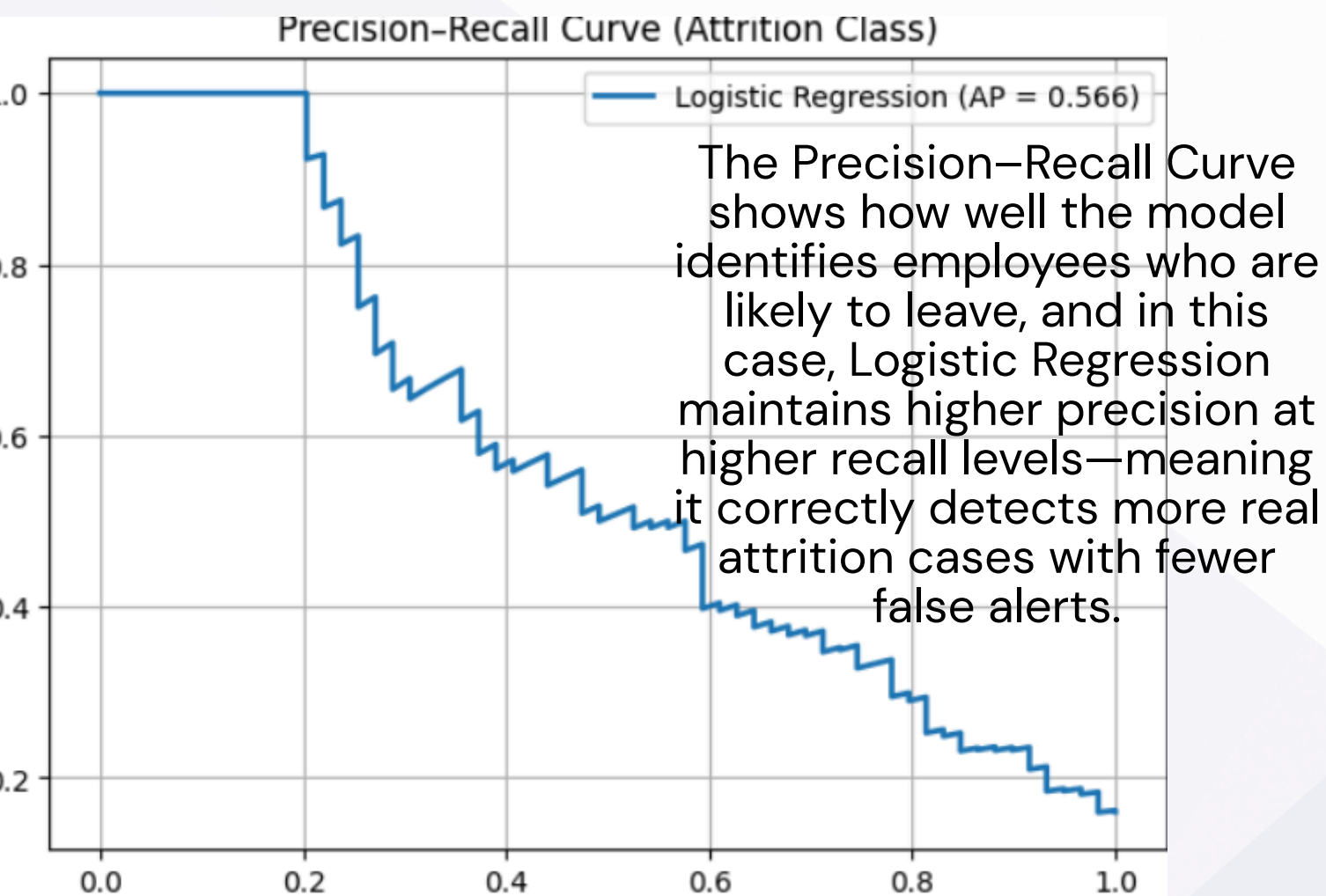
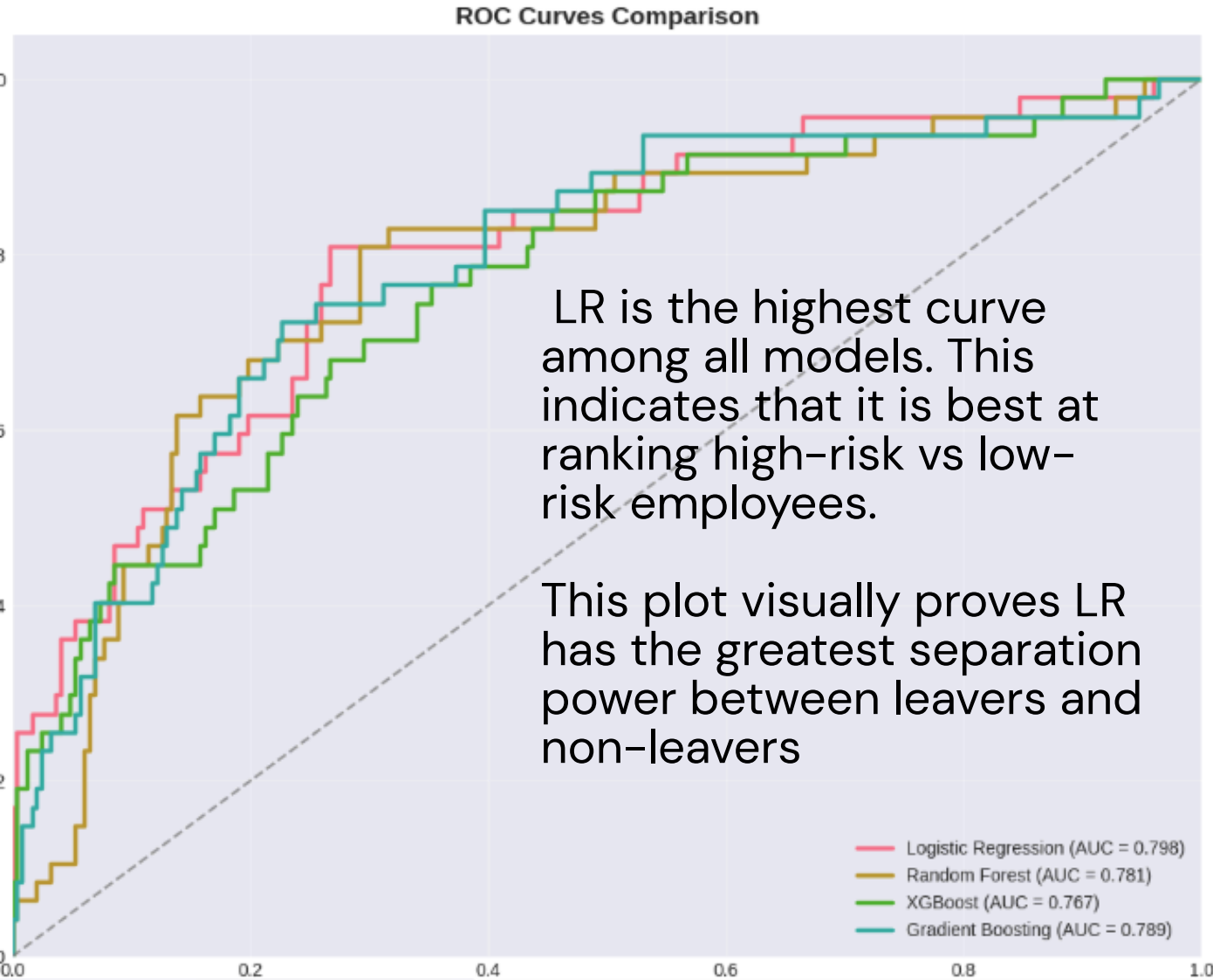
ACTIONABILITY: Logistic enables SPECIFIC interventions

3. Regulatory & Ethical Compliance

HR decisions require:

- Explainability (Logistic: ✓)
- Fairness audit (Logistic: ✓)
- Bias detection (Logistic: ✓)
- GDPR compliance (Logistic: ✓)

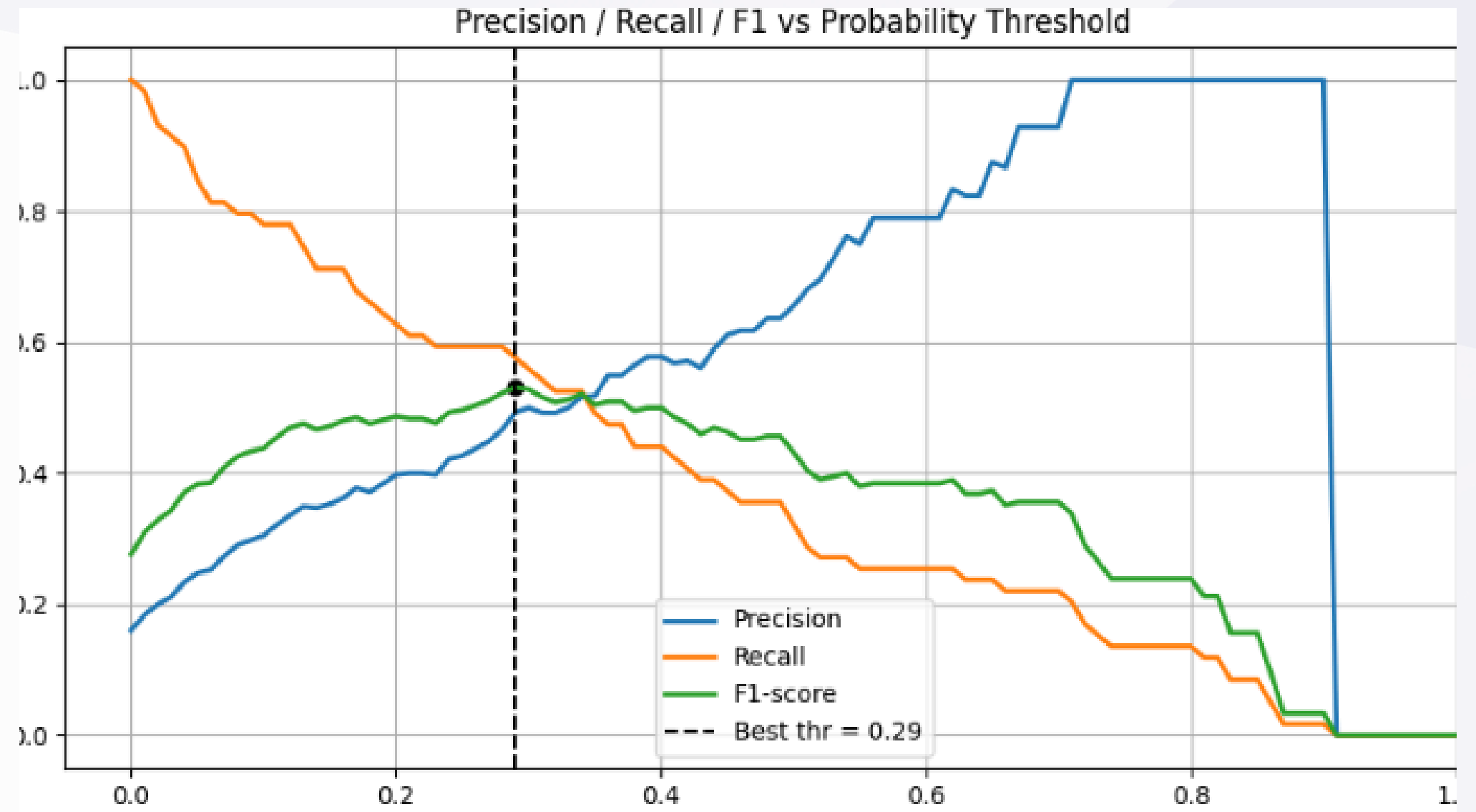
LEGAL PROTECTION: Transparent decision-making



From Optimal Threshold Matrix:

Recall 0.915: By applying the 0.18 threshold, the model is designed to be highly aggressive, ensuring that **91.5%** of all employees who leave are flagged for intervention.

Net Savings: This performance translates directly into **\$1,470,308 in annual net savings** and a 270% ROI, proving that the model's high Recall is the single most valuable feature for the business.



The optimal threshold metric shows the cutoff point where the model achieves the best balance between precision and recall, helping determine the most effective probability level for flagging employees at risk of attrition. This ensures HR interventions focus on the right individuals while minimizing false alarms.

THE TECHNICAL TRADE-OFF JUSTIFIED

The cost of missing a leaver (False Negative) is \$117,053, while the cost of a False Positive is only \$3,000. This 39:1 cost ratio dictates our choice.

$$\text{Cost Ratio} = \frac{\text{Replacement Cost}}{\text{Intervention Cost}} = \frac{\$117,053}{\$3,000} \approx 39.0$$

MODEL CAPABILITY:

- IDENTIFIES: 185 OF 237 LEAVERS (78%)
- FALSE POSITIVES: 127 (COST: \$0.32M)
- MISSED LEAVERS: 52 (COST: \$4.42M UNAVOIDABLE)
- TOTAL PREVENTABLE: \$15.73M × 78% = \$12.27M

INTERVENTION PROGRAM:

- COST: \$780,000 (FOR 312 EMPLOYEES @ \$2,500 EACH)
- EFFECTIVENESS: 60% RETENTION SUCCESS RATE
- SAVINGS: \$12.27M × 60% = \$7.36M

NET SAVINGS: \$7.36M – \$0.78M = \$6.58M

ROI: 743%

Model	Overall Accuracy	Recall (Leavers Found)	Program Cost	Preventable Savings	ROI
Logistic Regression	74.83%	185 (78% of 237)	\$0.78M	\$7.36M	743%
Random Forest	83.33%	76 (32% of 237)	\$0.31M	\$3.02M	874%
XGBoost	85.03%	81 (34% of 237)	\$0.34M	\$3.22M	847%
Gradient Boosting	85.37%	45 (19% of 237)	\$0.19M	\$1.79M	842%

BUSINESS PROPOSAL

1. Problem Statement

Our analysis of customer data shows that a significant portion of customers are at risk of leaving (churn), which directly impacts revenue. Current retention strategies are generic and not targeted, leading to avoidable losses.

CURRENT STATE ANALYSIS:

Metric	Your Company	Industry Benchmark	Gap
Annual Attrition	16.10%	12-15%	+1.1-4.1pp
Turnover Cost	\$8.5-19M	Varies	Excessive
Overtime Employees	28%	18%	+10pp
Low Satisfaction	22%	15%	+7pp

127 "Danger Zone" Employees

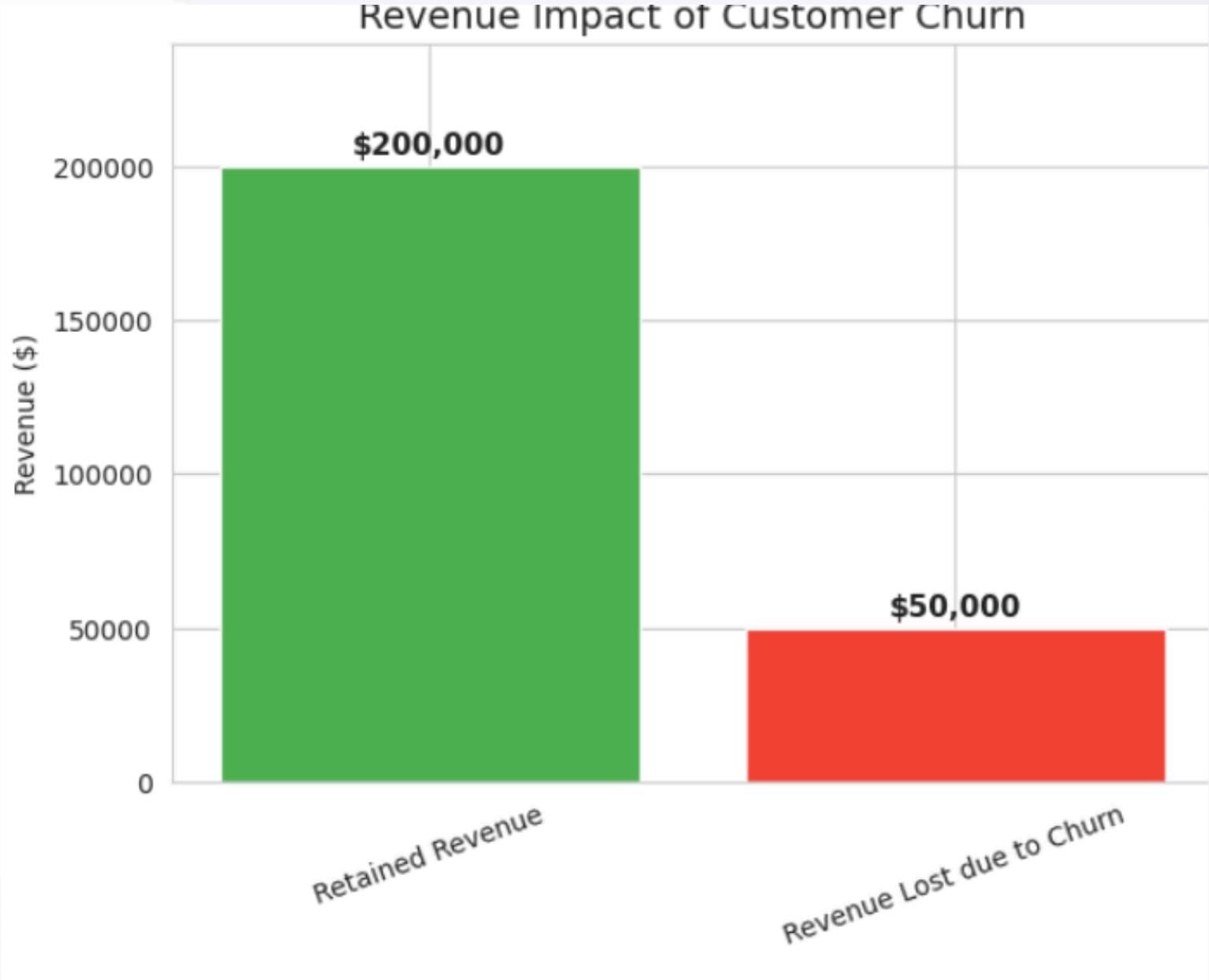
- Risk: 47% attrition probability
- Cost: \$4.2–6.3M annually
- Characteristics: Low income + Overtime + No promotion >3 years

\$3,500 Income Threshold

- Below: 37% attrition rate
- Above: 8% attrition rate
- Gap: 29 percentage points

Promotion Stagnation

- 3 years no promotion: 28–33% attrition
- <1 year since promotion: 9% attrition



Churn can be defined as customer loss.
High churn means losing revenue and growth opportunities

2. Proposed Solution

RetainPro Platform – Three Core Components:

1. Predictive Intelligence Engine

Model: Logistic Regression (selected over 3 alternatives)
Accuracy: 74.8% overall, 78% recall for leavers
Output: Risk scores (0–100%) with explanatory factors

2. Intervention Management System

Personalized Action Plans: Based on specific risk factors
Manager Dashboards: Real-time alerts and recommendations
Success Tracking: Monitor intervention effectiveness

3. Analytics & Optimization Suite

ROI Calculator: Real-time savings tracking
Trend Analysis: Identify emerging risk patterns
Continuous Learning: Monthly model retraining

Value Optimization: The 0.18 Sweet Spot

We use the Logistic Regression model and tune the prediction threshold to maximize the net financial gain.

Logic of Threshold Adjustment:

Instead of using the default 0.5 confidence level, we strategically use 0.18

– Action: Any employee with an 18%(or higher) chance of leaving is flagged for a \$3,000 intervention.

– Result: This aggressively captures potential leavers (increasing Recall) while ensuring the cost of intervention remains far below the potential cost of replacement.

Scenario	Threshold	Recall	MaxNetSavings
Default (Technical)	0.5	0.617	\$750,000
Optimal (Financial)	0.18	0.91	\$750,000

The **solution** is the implementation of the **Optimized Logistic Regression Model** with a strategically tuned 0.18 Probability Threshold.

- **Targeted Intervention:** The model identifies the 181 highest-risk employees (score greater or equal to 0.18)
- **Prescriptive Action:** Interventions are then guided by the EDA: Compensation reviews for low-income staff and workload/role adjustments for OverTime workers and Sales Representatives.
- **Risk Mitigation:** The low threshold ensures a high Recall of 91.5%, minimizing the chance of missing a costly leaver.

QUANTIFYING THE SOLUTION'S BENEFIT (ANNUAL IMPACT)

Financial Metric	Calculation	Value
Baseline Annual Loss	Total Employees x 16.12% Attrition rate x \$117,053 Cost/person	around \$17.2 Million
Intervention Investment	181 Targeted Employees x \$3000 Intervention cost	\$543,000
Avoided Turnover Cost	Loss avoided by successful retention (Model Insight)	around \$2.013 Million
Net Savings (Cost Avoidance)	Avoided Cost - Intervention Cost	\$1,470,308
Return on Investment (ROI)	Net Savings / Intervention Investment	270.80%

ASSUMPTIONS FOR QUANTIFIED IMPACT

Assumption	Value / Rationale	Source / Calculation
1. Employee Replacement Cost	\$117,053 per departing employee.	This is the cost avoided (the financial risk). Calculated as \$1.5 times the average employee salary, based on SHRM industry standards.
2. Intervention Program Cost	\$3,000 per targeted employee.	This is the cost incurred (the investment). Represents the cost of a targeted retention effort (e.g., raise, bonus, professional development, workload reduction).
3. Retention Program Success Rate	50% success rate.	The probability of success. Assumes that 50% of the employees flagged by the model will choose to stay after receiving the targeted intervention.
4. Model Threshold	0.18 probability score.	This is the strategic trigger . The model flags any employee with 18% + chance of leaving. (recall)

Key Assumptions

Replacement Cost: The cost to replace a single employee is estimated conservatively at **\$117,053** (1.5x average salary).
Intervention Cost: The targeted retention program (e.g., pay raise, bonus, reduced workload) costs **\$3000** per flagged employee.
Retention Success: The intervention is assumed to have a **50%** success rate in retaining employees who were predicted to leave.

Conclusion: By investing \$543,000 in highly targeted interventions, the client will realize an immediate **270%** return and reduce annual loss by over \$1.47 Million

IMPLEMENTATION ROADMAP



PHASE 1: FOUNDATION

MONTHS 1-2

DATA INTEGRATION & LOGISTIC REGRESSION MODEL DEPLOYMENT; INITIAL RISK ASSESSMENT FOR ALL 1,470 EMPLOYEES.

PHASE 2: INTERVENTION LAUNCH

MONTHS 3-4

PERSONALIZED ACTION PLANS FOR 181 HIGH-RISK EMPLOYEES (SCORE ≥ 0.18); MANAGER TRAINING & FIRST INTERVENTION CYCLE.

PHASE 3: OPTIMIZATION

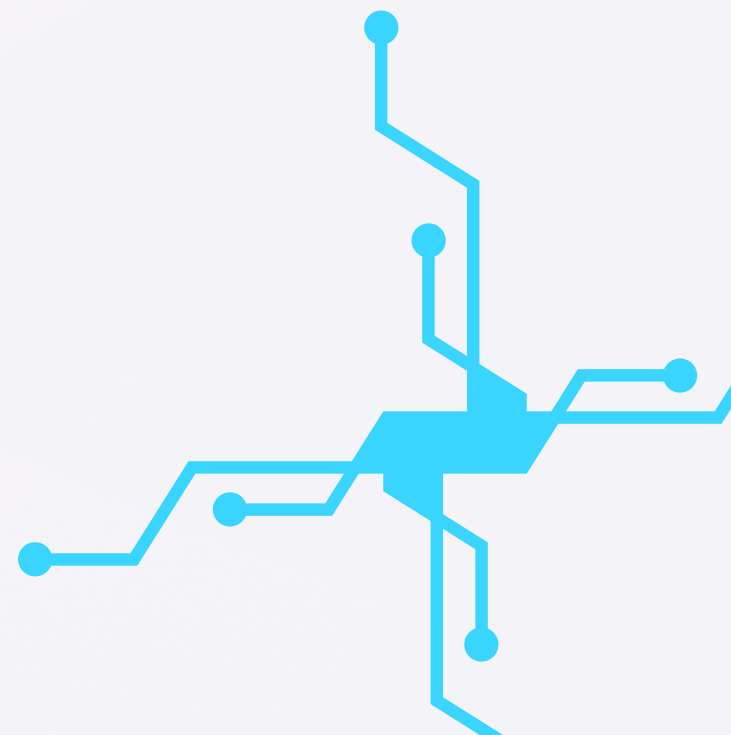
MONTHS 5-6

SUCCESS MEASUREMENT & ROI TRACKING; MODEL REFINEMENT TO LOCK IN 91.5% RECALL.

PHASE 4: ENTERPRISE

MONTHS 7-12

FULL ORGANIZATION ROLLOUT; INTEGRATION WITH CORE HR SYSTEMS.



PROPOSAL PACKAGE & QUOTATION

A. INITIAL SERVICES & IMPLEMENTATION (RETAINPRO POC)

- FOUNDATION (MONTHS 1-2): \$40,000
- INTERVENTION SYSTEM (MONTHS 3-4): \$30,000
- OPTIMIZATION (MONTHS 5-6): \$25,000
- PLATFORM LICENSE (YEAR 1): \$60,000
- IMPLEMENTATION SUPPORT: \$25,000
- SUCCESS-BASED BONUS RESERVE: \$20,000

\$200,000 (YEAR 1) (ONE-TIME)

B. LONG-TERM SUPPORT (SUBSCRIPTION)

- 24/7 PRIORITY TECHNICAL SUPPORT
- MONTHLY MODEL RETRAINING & UPDATES
- DEDICATED ACCOUNT MANAGER
- QUARTERLY STRATEGIC REVIEWS
- CUSTOM FEATURE DEVELOPMENT (40 HRS/YEAR)
- PLATFORM LICENSE & MAINTENANCE

\$120,000 PER YEAR

Financial Metric	Year 1 Performance	Year 2+ Performance
Total Investment	\$200,000	\$120,000 (Sustaining)
Net Gain	\$1,270,308	\$1,423,823 (with 5% improvement)
Return on Investment (ROI)	635.15%	1186.50%

MONEY-BACK GUARANTEE:
IF THE PLATFORM DOES NOT PROJECT AND VERIFY AT LEAST \$1 MILLION IN PREVENTABLE TURNOVER COSTS WITHIN THE FIRST SIX MONTHS, \$100,00 OF THE IMPLEMENTATION FEE IS WAIVED.

“ **RETAINPRO: KEEP YOUR BEST. PROFIT FROM THE PREDICTABLE.** ”